

Markscheme

Mathematics: analysis and approaches

Practice question bank

20. (a)

Consider the function $f(x) = \frac{3x - 1}{(x - 2)^2}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x - 2} + \frac{B}{(x - 2)^2}$.

attempt to write $3x - 1 = A(x - 2) + B$ (M1)

substituting $x = 2$: $5 = B$ A1

comparing x coefficients (M1)

$3 = A$ A1

[4 marks]